

PolyFitFX Application - User Manual

Table of Contents

1. Introduction
 2. System Requirements
 3. Installation
 4. Getting Started
 5. Importing Data
 6. Fitting a Polynomial
 7. Understanding the Output
 - Model Metrics
 - Regression Plot
 - Residual Plot
 - Confidence Interval
 8. Adding and Removing Data Points
 9. Exporting Data
 10. Troubleshooting
 11. Frequently Asked Questions
 12. Contact and Support
-

1. Introduction

Welcome to the Polynomial Regression Application!

This tool allows you to visualize and analyze data using polynomial regression. You can import data from CSV files, fit polynomials of any degree, view model metrics (R^2 , RMSE, MAE), see residual plots, and interact with the results.

2. System Requirements

- Java 21
- JavaFX 23.0.1
- Apache Commons Math3 library (included with the application)
- Windows, macOS, or Linux

3. Installation

1. Download the application JAR file or build from source.
2. Ensure Java SDK 21 is installed on the system.
3. Double-click the JAR file or run it from the command line:

```
java -jar PolynomialRegressionApp.jar
```

4. Getting Started

When you launch the application, you'll see:

- A menu bar at the top
- Controls for selecting polynomial degree and fitting the model
- A table for data points
- A main chart for regression results
- A residual plot
- A metrics display area

5. Importing Data

1. Click File > Import CSV.
2. Select a CSV file with two columns (no header), e.g.:

```
1.0,2.5
```

```
2.0,3.7
```

```
3.0,5.1
```

3. The data will appear in the table and be plotted on the chart.

Tips:

- The file should not have a header row.
- Each line should be: x-value, y-value

6. Fitting a Polynomial

1. Enter the desired polynomial degree (e.g., 2 for quadratic) in the degree field.
2. Click the 'Fit Polynomial' button.
3. The app will fit the model and update the charts and metrics.

7. Understanding the Output

Model Metrics

- R^2 (Coefficient of Determination): Indicates how well the model fits the data (1.0 = perfect fit).
- RMSE (Root Mean Square Error): Average prediction error magnitude.
- MAE (Mean Absolute Error): Average absolute prediction error.

Regression Plot

- Shows your data points and the fitted polynomial curve.
- Confidence bands (shaded area or dashed lines) indicate uncertainty.

Residual Plot

- Plots residuals (errors) vs. fitted values.
- Helps diagnose model fit (look for random scatter).

8. Adding and Removing Data Points

- Add: Enter X and Y values in the fields below the controls and click Add Point.
- Remove: Select a row in the data table and click Remove Selected.

The charts and metrics update automatically.

9. Exporting Data

- Export CSV:
Use File > Export CSV to save your current data points to a CSV file.

10. Troubleshooting

- Error importing CSV: Make sure your file has no header and only numeric values.
- Blank charts or metrics: Ensure you have loaded data and specified a valid degree.
- App won't start: Check your Java installation and ensure all dependencies are present.

11. Frequently Asked Questions

Q: Can I use a CSV with a header row?

A: No, the app expects no header. Remove any header before importing.

Q: What does the confidence interval mean?

A: It shows the range where the true regression curve is likely to fall, given the data.

Q: How do I reset the app?

A: Close and reopen the application, or clear all data points manually.